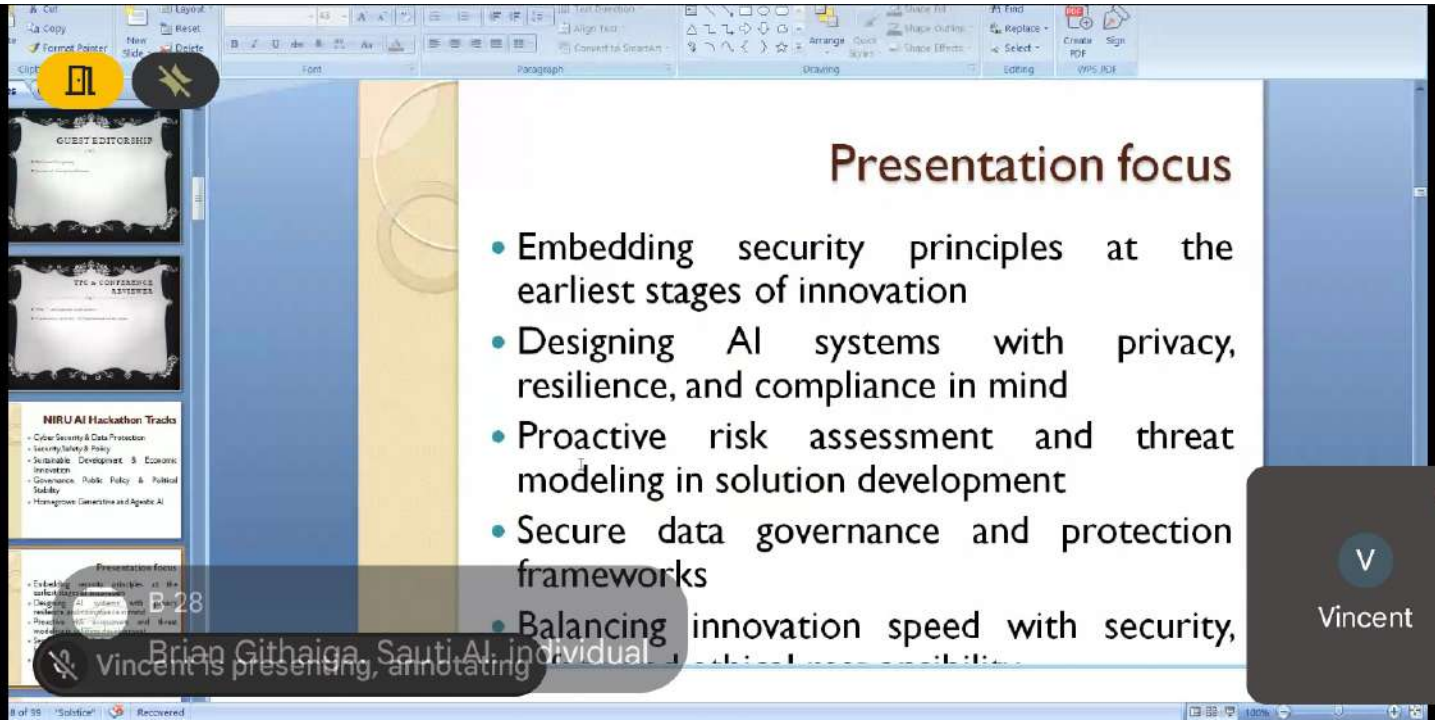
Presentation - Microsoft PowerPoint
NATIONAL DEFENCE UNIVERSITY-KENYA
Innovation with Security by Design
Dr Vincent Nyangaresi, PhD
Jaramogi Oginga Odinga University of Science and Technology
Vincent is presenting, annotating
mahmoud


Presentation focus
Embedding security principles at the earliest stages of innovation
Designing AI systems with privacy, resilience, and compliance in mind
Proactive risk assessment and threat modeling in solution development
Secure data governance and protection frameworks
Balancing innovation speed with security,
GUEST EDITORSHIP
NIRU AI Hackathon Tracks
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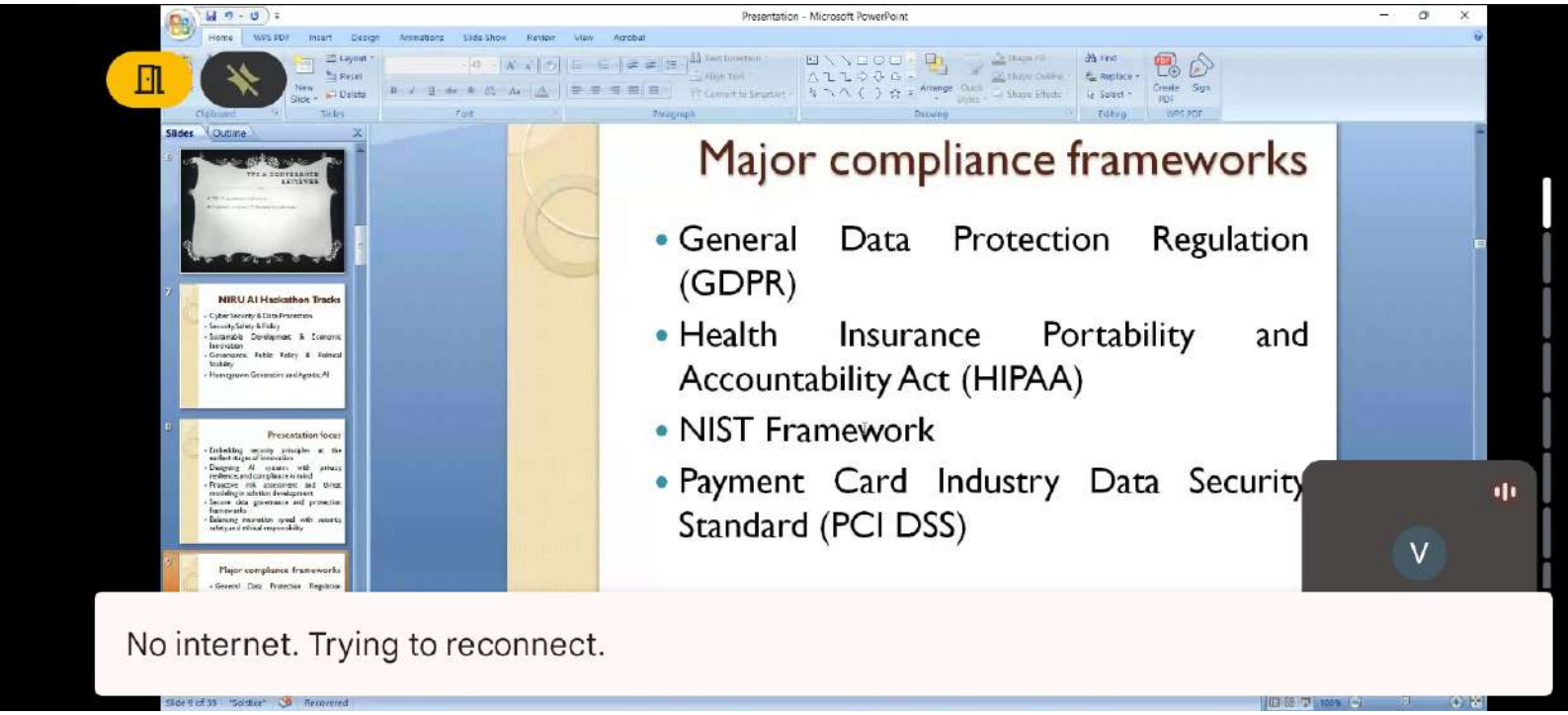
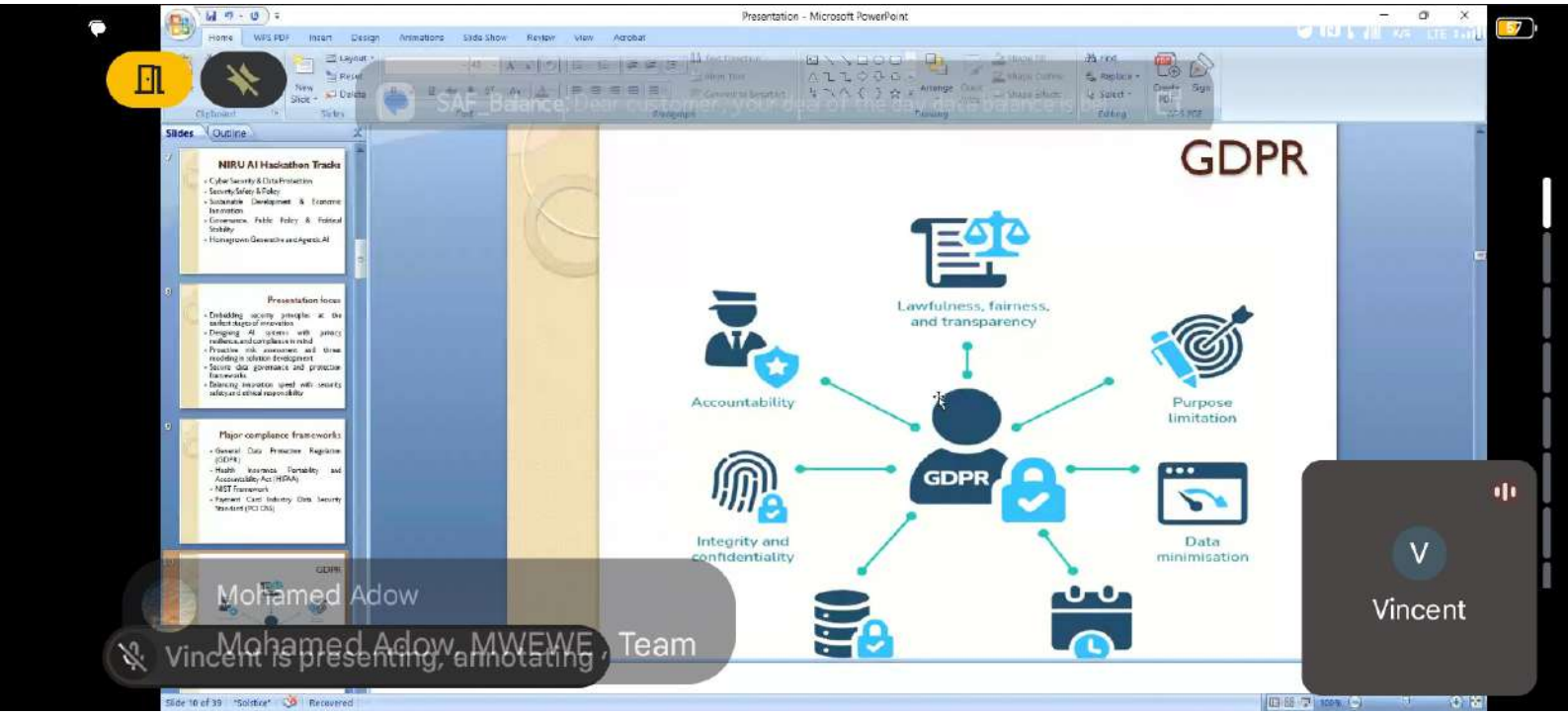
Brian Githaiga, Sauti AI, individual


Presentation - Microsoft PowerPoint
Major compliance frameworks
General Data Protection Regulation (GDPR)
Health Insurance Portability and Accountability Act (HIPAA)
NIST Framework
Payment Card Industry Data Security Standard (PCI DSS)
No internet. Trying to reconnect.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GDPR
Lawfulness, fairness, and transparency
Accountability
Purpose limitation
GDPR
Integrity and confidentiality
Data minimisation
Mohamed Adow
Mohamed Adow, MWEWE , Team
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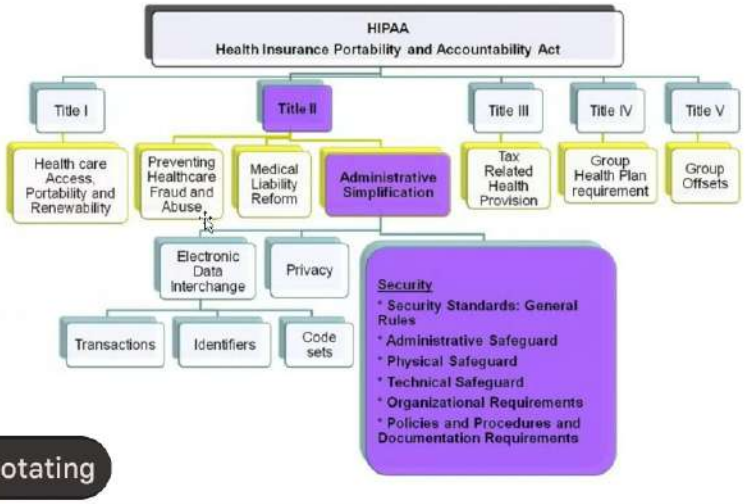
HIPAA



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NIST Cyber Security Framework

Identify	Protect	Detect	Respond	Recover
Asset Management	Access Control	Anomalies and Events	Response Planning	Recovery Planning
Business Environment	Awareness and Training	Security Continuous Monitoring	Communications	Improvements
Governance	Data Security	Detection Processes	Analysis	Communications
Risk Assessment	Info Protection Processes and Procedures		Mitigation	
Risk Management Strategy	Maintenance		Improvements	
	Protective Technology			

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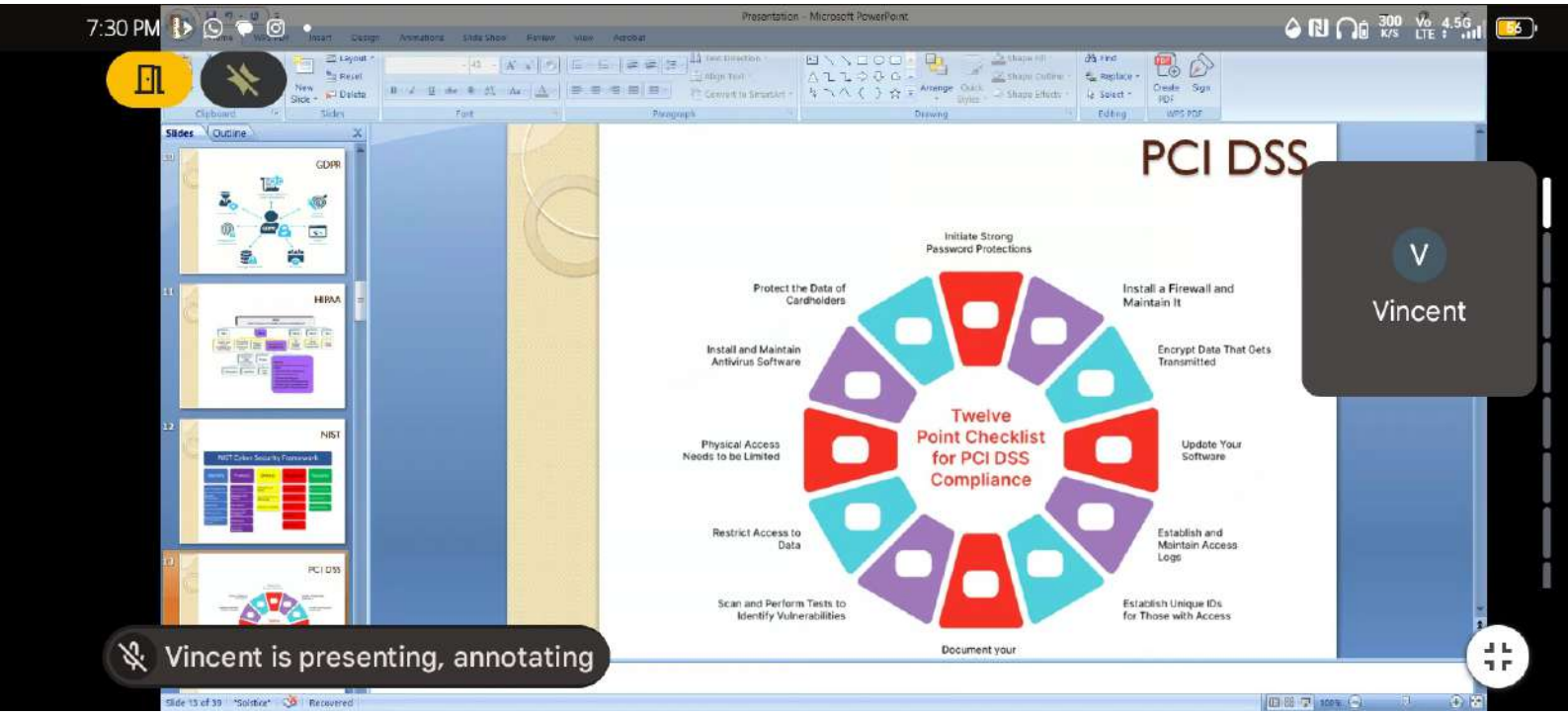
7:30 PM
Presentation - Microsoft PowerPoint
Slides
Outline
GDPR
HIPAA
NIST
PCI DSS
PCI DSS
Initiate Strong Password Protections
Install a Firewall and Maintain It
Encrypt Data That Gets Transmitted
Update Your Software
Establish and Maintain Access Logs
Establish Unique IDs for Those with Access
Document your
Scan and Perform Tests to Identify Vulnerabilities
Restrict Access to Data
Physical Access Needs to be Limited
Install and Maintain Antivirus Software
Protect the Data of Cardholders
Twelve Point Checklist for PCI DSS Compliance
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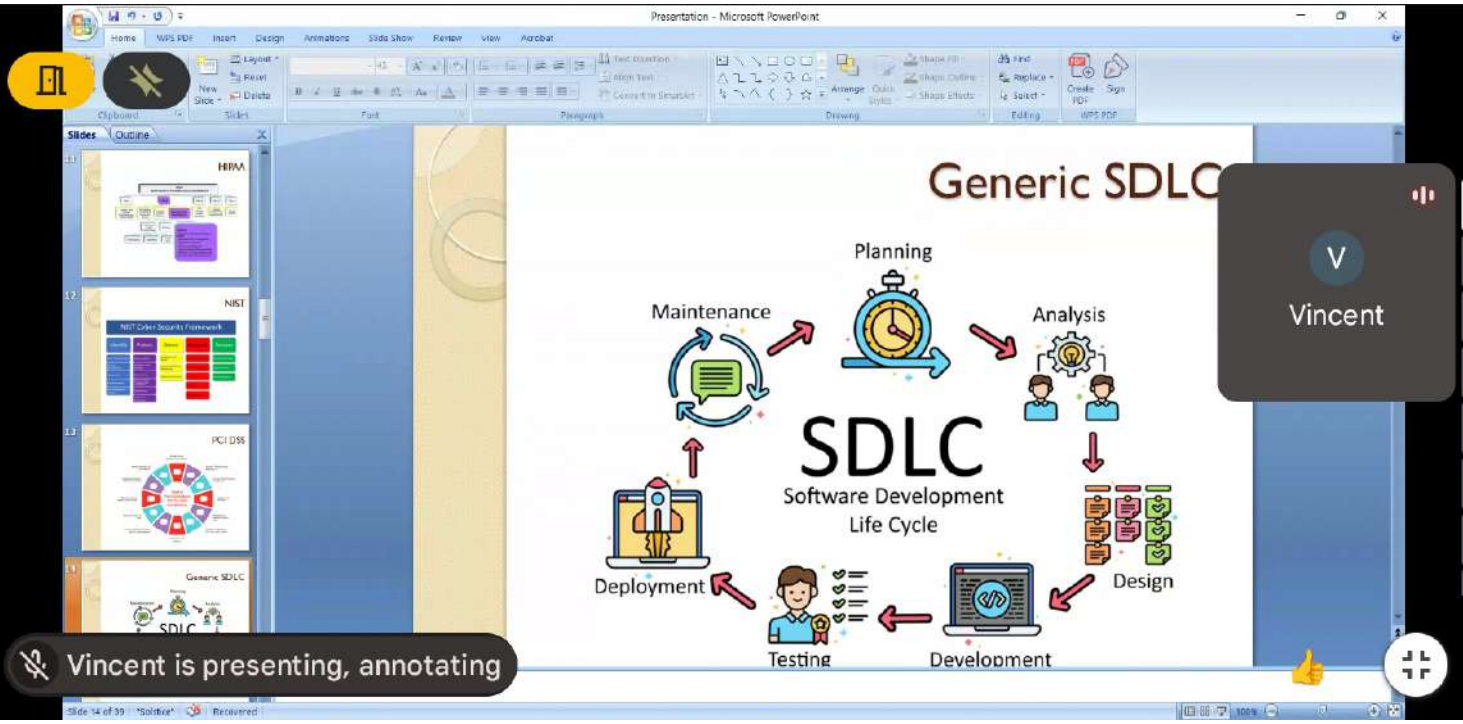
Presentation - Microsoft PowerPoint
Generic SDLC
Planning
Analysis
Design
Development
Testing
Deployment
Maintenance
SDLC
Software Development
Life Cycle
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7:33 PM
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Rewel +107
Secure Software Development
Lifecycle (SSDL
Risk Assessment
Threat Modelling &
Design Review
Requirements
Design
Deployment
Development
Static Analysis
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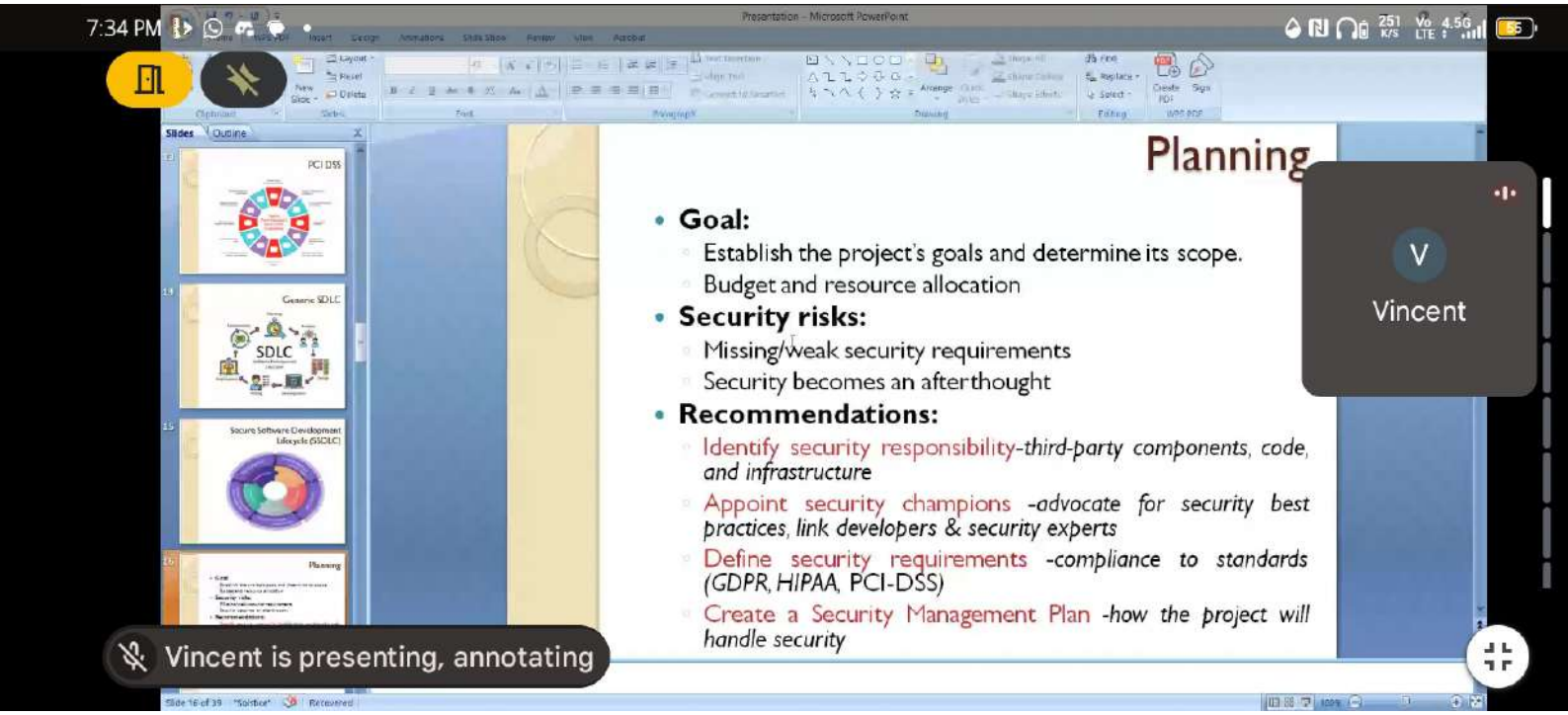
7:34 PM
Presentation - Microsoft PowerPoint
Slides
Outline
PCI DSS
Generic SDLC
SDLC
Secure Software Development Lifecycle (SSDLC)
Planning
Planning
Goal:
Establish the project's goals and determine its scope.
Budget and resource allocation
Security risks:
Missing/weak security requirements
Security becomes an afterthought
Recommendations:
Identify security responsibility-third-party components, code, and infrastructure
Appoint security champions -advocate for security best practices, link developers & security experts
Define security requirements -compliance to standards (GDPR, HIPAA, PCI-DSS)
Create a Security Management Plan -how the project will handle security
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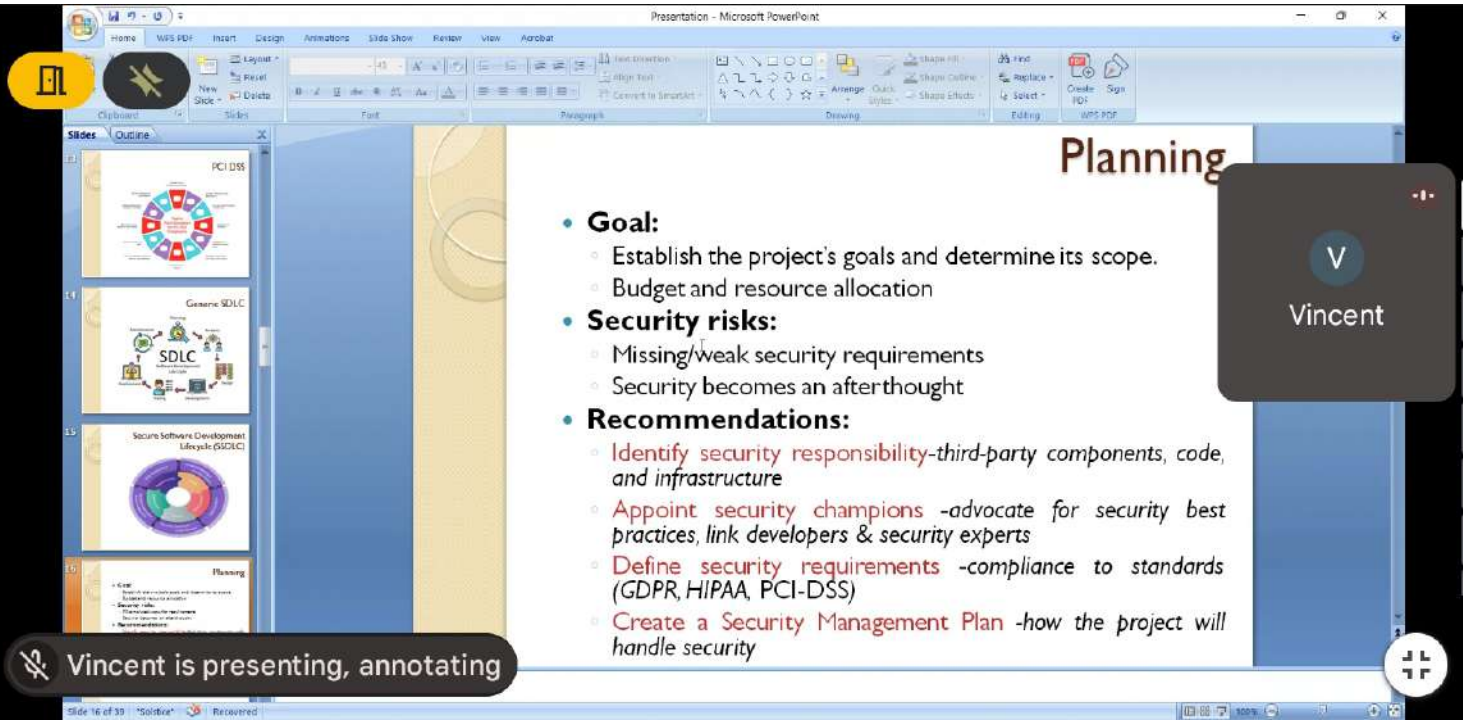
Vincent is presenting, annotating


Presentation - Microsoft PowerPoint
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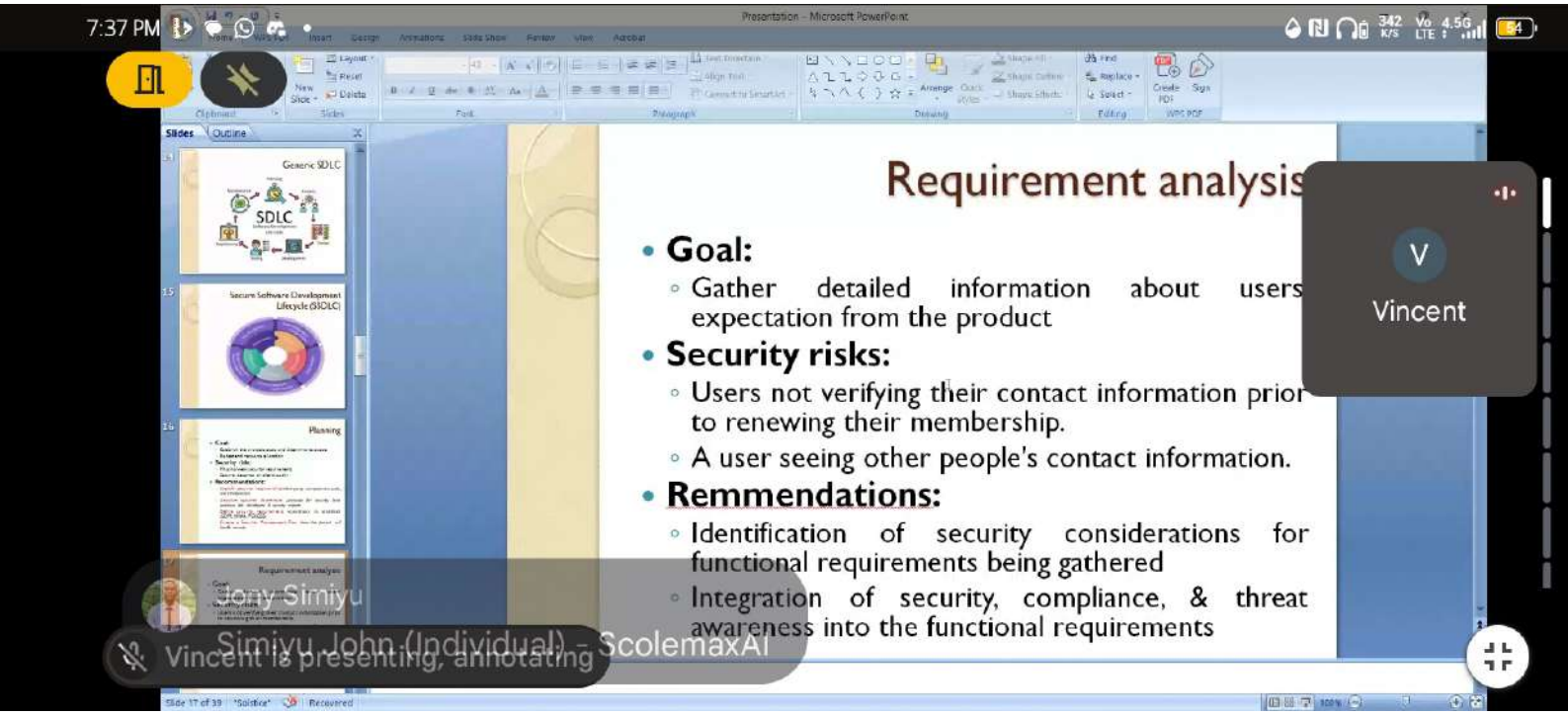
7:37 PM
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Requirement analysis
Goal:
Gather detailed information about users expectation from the product
Security risks:
Users not verifying their contact information prior to renewing their membership.
A user seeing other people's contact information.
Remmendations:
Identification of security considerations for functional requirements being gathered
Integration of security, compliance, & threat awareness into the functional requirements
Vincent
Jony Simiyu
Simiyu, John (Individual) - ScolemaxAI
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7:40 PM

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Design

- **Goal:**
  - Develop the product/software's architecture, e.g data flow diagrams, mock-ups of user interfaces
- **Security risks:**
  - Users retrieving information from the database devoid of a valid session token
- **Recommendations:**
  - Threat modeling -*early identification of vulnerabilities*
  - Application of the principle of least privilege
  - Implementation of defense-in-depth
  - Integration of security requirements into the architecture

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Raymond Ouko

C

You

107 others

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7:43 PM

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Development

- **Goal:**
  - Write the actual code for the product/software
- **Security risks:**
  - Using open source libraries devoid of checking them for vulnerabilities
  - Processing data devoid of validating user inputs
- **Recommendations:**
  - Secure coding guidelines/standards
  - Code reviews
  - Software Composition Analysis (SCA) tools-*for checking open-source components*

d notes

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Testing

- **Goal:**
  - Thorough checks to identify bugs/ issues within the product/software
  - Confirm that everything is working properly:
    - Unit testing
    - Security testing
    - Integration testing
- **Security risks:**
  - Failure to check the critical paths of your application
  - Failure to verify the correctness of the underlying application
- **Recommendations:**
  - Integration of automated vulnerability scanning
  - Manual penetration testing
  - Proactive DevSecOps approach
  - Use automated tools such as CI/CD pipelines- *to control verification and release*

d notes

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Trying to speak? Turn on your microphone.

7:50 PM

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Deployment

- **Goal:**
  - Launching the product/software to the end-users.
- **Security risks:**
  - Mis-configured environments
  - Compromised third-party dependencies
  - Insufficient environment segregation
- **Repercussions:**
  - Unauthorized code changes
  - Data breaches
  - System instability
- **Recommendations:**
  - Implement Role-Based Access Control (RBAC)-principle of least privilege to deployment tools
  - Sign and Verify Artifacts
  - Harden Pipeline Infrastructure-no lateral movement

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Maintenance

- **Goal:**
  - Addressing user feedback, ongoing support and regular updates to the product/software
- **Security risks:**
  - Zero-day vulnerabilities
  - External penetration tests
- **Recommendations:**
  - Ongoing vulnerability management
  - Regular patching
  - Monitoring & detection

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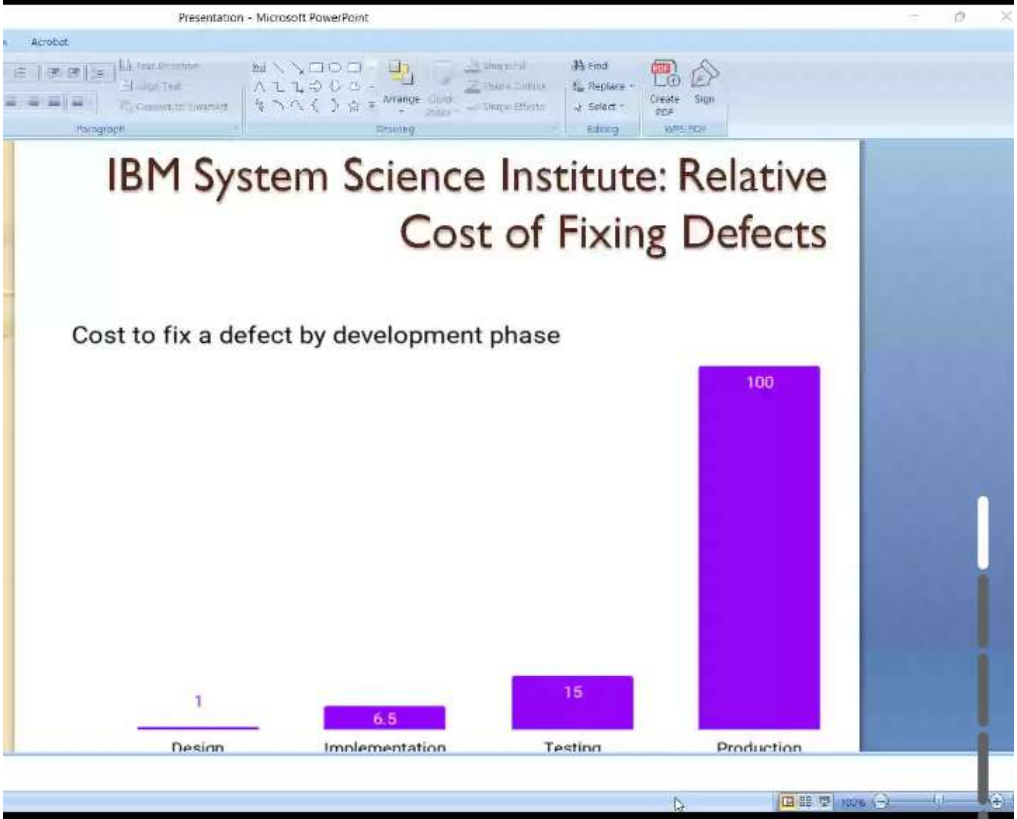

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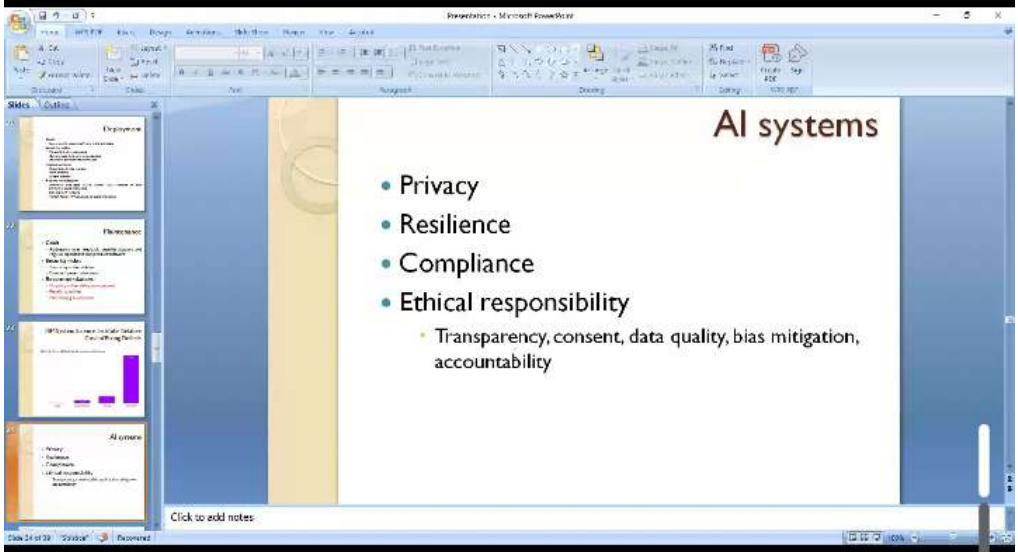

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Several participants left the meeting

Social Media

Algorithms analyze user behavior for ads but risk spreading bias and misinformation.

Facial Recognitio

Captures facial data identification, creat serious privacy an surveillance concer

4

Voice Assistants

Devices record conversations for processing, potentially storing sensitive and

5

Web Activity Monito

Tracks browsing hab personalize conte risking user data pri

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AI & privacy risks: examples and implications

Unintentional Bias

Imbalanced training data may cause AI to produce unfair or discriminatory results, affecting people and shaping decisions.

Inaccurate Predictions

Faulty predictions in fields like healthcare or finance can lead to serious misdiagnoses or financial loss.

Data Privacy Breach

Poor data security can expose personal information, leading to breaches and privacy violations.

Unforeseen Consequences

When complex AI models fail or produce inaccuracies, they can cause unexpected effects across industries and systems.

Job Displacement

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AI in healthcare: security & Privacy concerns

Security Risks

Data Breaches
AI processes vast amounts of sensitive data, increasing the risk of cyberattacks on healthcare systems.

AI System Vulnerabilities
Hackers can manipulate AI inputs, leading to incorrect diagnoses or treatments.

Ransomware Risks
Ransomware attacks can paralyze AI systems, disrupting patient care and access to critical data.

Privacy Concerns

Data Anonymization Challenges
Anonymized patient data can still be re-identified, compromising privacy.

Consent and Data Ownership
Patients may not fully understand AI data use, and unclear ownership raises legal issues.

Algorithm Bias
AI trained on biased data can produce unfair or discriminatory outcomes.

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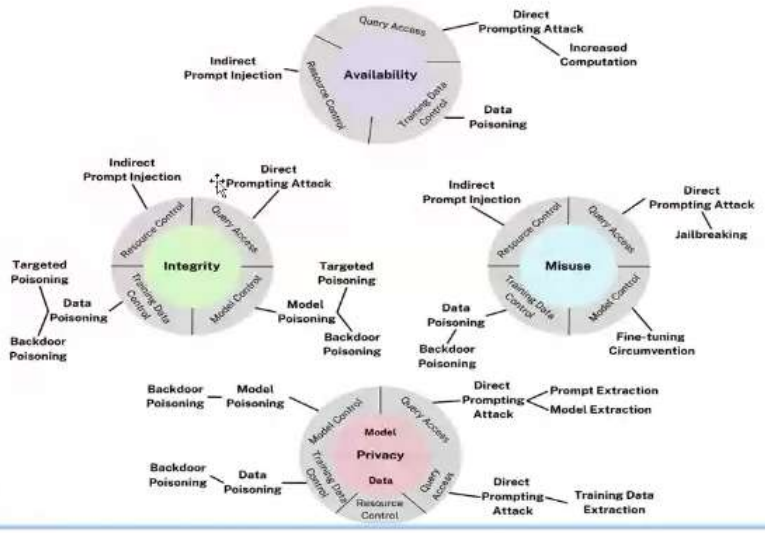
AI security and AI safety risks



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Way forward

Best practices for ethical AI implementation

1. Data Minimization
2. Anonymization and Privacy Protection
3. Bias Detection and Fairness Testing
4. Regular Audits and Continuous Monitoring
5. Stakeholder Engagement and Inclusivity
6. Human Oversight and Accountability

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...Continued

Privacy by design in AI development

1. Conduct privacy impact assessments early
2. Apply strict data minimization
3. Use anonymization and protective techniques
4. Leverage federated learning
5. Form ethics and privacy review committees
6. Automate consent management

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GDPR & AI Compliance

Data Mapping
Need to identify personal data, data flows, data subjects, and data recipients within the organization's AI ecosystem.

Privacy Impact Assessment
Include risk identification, risk evaluation, and risk mitigation strategies.

Legal Basis & Consent
Clear consent in AI applications while ensuring compliance with GDPR requirements.

Transparent Communication
Need to provide information on the purpose, processing activities, data recipients, and data subject rights.

Data Subject Rights
Need to implement mechanisms for data subjects to exercise their rights within AI systems.

Staff Training & Awareness
Need for awareness programs to keep employees informed about evolving AI and GDPR landscape.

Incident Reponse
Importance of having a strong incident response plan to meet GDPR's breach reporting requirements.

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8:20 PM

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Balancing innovation speed with security, safety, and ethical responsibility

- Also referred to as 'Responsible AI' or 'Ethics by Design'
- **Core Strategies:**
  - Embed ethics by design-*Integrate ethical checks (e.g, privacy impact assessments, bias audits) in SDLC*
  - Determine governance frameworks - *Ethics board to ensure accountability & guidelines (e.g.,NIST AI Risk Management Framework)*
  - Security as a foundation - *e.g, pen testing, data anonymization*
  - Transparency & explainability - *AI decisions should be understandable to boost user trust & regulatory*

Click to add notes

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Is this a fail-safe?

Connect to Server

10.1.1.202 test failed.
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 2026 (HY000): SSL connection error: error:1425F102:SSL routines:ssl_choose_client_version:unsupported protocol

OK

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Abstract

Document Sections

I. Introduction

II. Attack Model

III. Review of Sibahee Et Al.'s Protocol

IV. Security Analysis of Sibahee Et Al.'s Protocol

V. Security Analysis

Show Full Outline

Authors

Abstract:
In vehicular Ad Hoc networks (VANETs) environments, information needs to be exchanged frequently between vehicles, road side units (RSUs), and management centers to support safe and efficient traffic management. However, due to the openness of wireless communication and dynamic network topology, VANETs face many security threats, such as forging, eavesdropping, and man-in-the-middle attacks. Therefore, it is crucial to design efficient and secure authentication protocols. To address this problem, Sibahee et al. proposed a two-factor privacy-preserving authentication protocol in IEEE Internet of Things Journal (pp. 14253–14266, DOI: 10.1109/JIOT.2023.3340259, April 15, 2024) and claimed that it can resist multiple attacks. However, we find that the protocol cannot resist temporary random number leakage attacks and privileged insider attacks, and does not have perfect forward security. To address the above problems, this article proposes an improved authentication scheme based on elliptic curve cryptography (ECC) to enhance the security of the scheme. We conduct formal (real-or-random Model) and informal security analyses of the improved scheme and evaluate its performance in terms of computational overhead and communication cost. The experimental results show that the improved scheme has significant advantages in both security and performance.

Published in: IEEE Internet of Things Journal (Volume: 12 , Issue: 16, 15 August 2025)

Elliptic curve cryptography based mutual authentication protocol for low computational capacity RFID systems - performance analysis by simulations
2010 IEEE International Conference on Wireless Communications, Networking and Information Security
Published: 2010

Anonymous Authentication Protocol Based on Physical Unclonable Function and Elliptic Curve Cryptography for Smart Grid
IEEE Systems Journal
Published: 2025

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